

JNTU Hyderabad

REGULAR EXAMINATION ? 2025

Heat Transfer ? Regular Examination, 2025

Subject Code: ME402 Semester: 4 Regulation: R20

Time: 3 Hours Max. Marks: 70

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.
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SECTION A (Unit 1: Conduction Heat Transfer)

1. Explain Conduction Heat Transfer with a neat diagram/sketch and a suitable example. [12 Marks]
2. Write short notes on Conduction Heat Transfer. [12 Marks]
3. Define Conduction Heat Transfer. Explain its significance in Heat Transfer with a suitable example. [12 Marks]

SECTION B (Unit 2: Convection Heat Transfer)

4. Write short notes on Convection Heat Transfer. [12 Marks]
5. Describe the working principle of Convection Heat Transfer with necessary equations. [12 Marks]
6. Solve a problem related to Convection Heat Transfer. Show all the steps clearly. [12 Marks]

SECTION C (Unit 3: Radiation Heat Transfer)

7. Solve a problem related to Radiation Heat Transfer. Show all the steps clearly. [12 Marks]
8. Discuss the engineering applications of Radiation Heat Transfer. [12 Marks]
9. Solve a problem related to Radiation Heat Transfer. Show all the steps clearly. [12 Marks]

SECTION D (Unit 4: Heat Exchangers)

10. Solve a problem related to Heat Exchangers. Show all the steps clearly. [12 Marks]
11. List the advantages and limitations of Heat Exchangers. Explain any two in detail. [12 Marks]
12. Define Heat Exchangers. Explain its significance in Heat Transfer with a suitable example. [12 Marks]

SECTION E (Unit 5: Boiling and Condensation)

13. Describe the working principle of Boiling and Condensation with necessary equations. [12 Marks]
14. Discuss the engineering applications of Boiling and Condensation. [12 Marks]
15. Solve a problem related to Boiling and Condensation. Show all the steps clearly.

[12 Marks]

*** End of Question Paper ***